Predictable Sparse and Spectral Contrast Monitoring of Real QEC Syndromes: A Predeclared Benchmark with a Disclosed Post-Detector Repair

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Abstract

Quantum-error-correction (QEC) records can be nonstationary and burst-prone, but detector firing rate is already a strong diagnostic and elaborate feature maps need not improve operational decisions. We introduce S-PACE, an initially preregistered composite of past-only sparse and spectral contrast witnesses for classical syndrome streams. Bounded predictable scores are converted to betting factors and combined with fixed priors over features, time scales, bet sizes, and 51 round-role experts. One complete paired shot—not one role—is the formal time step. Exact false-alarm validity is reserved for an independently randomized complete-pair design; the natural Google replay and the constructed PNNL/IBM snapshot boundaries are evaluated empirically. The locked benchmark compares S-PACE with paired detector firing rate, within-shot Page–CUSUM, diagonal detector surprise, shrinkage Hotelling monitoring, and an online logistic witness receiving the same feature bank. Detector scores were frozen under the original pre-access chain. After that detector run, a source-derived consumer-validation repair was separately ratified before decoder outcomes, PNNL held payloads, or a completed randomization replicate were available. Detector numeric diagnostics had been exposed during diagnosis but were not used to select the repair, and the detector was neither modified nor rerun. After the locked outcome stage completed, a separate schema mismatch stopped the first strict publication extraction. Its publication-consumer-only repair changed no upstream evidence, gate, or recorded runtime and is not described as outcome-blind. A conjunctive two-arm rule prevents a favorable auxiliary analysis from rescuing a failed primary test. `TBD(CLAIM_SENTENCE)`

Keywords: quantum error correction; sequential change detection; e-processes; preregistration; sparse features; spectral contrasts

1 Introduction

Repeated syndrome measurements serve two distinct functions in QEC: they support decoding, and they expose changes in the physical error process. Detector rates, detector correlations, and occasional bursts are established hardware diagnostics [1, 2]. Modern work also adapts noise estimates and decoding graphs under drift [3–5]. A new monitor therefore earns its complexity only if it improves a fixed operational endpoint at matched observations, calibration, and alert budget.

The motivating idea is to follow coordinates or operator directions with large *between-stream contrast*, rather than directions selected only for pooled variance. Sparse discriminative projections and tailored high-dimensional change monitors already embody this broad principle [6]. Likewise, predictable betting, fixed mixtures of e-values, Page–CUSUM, Hotelling statistics, and the Jordan positive-part construction are established foundations [7–12]. The contribution pursued here is their auditable integration into an initially preregistered real-QEC protocol, followed by a disclosed post-detector, pre-outcome validator amendment:

1. a fixed raw-check plus pair-correlation feature bank and a positive-semidefinite spectral representation;
2. past-only sparse and spectral witnesses whose current scores remain bounded;
3. a complete-shot mixture that does not count 51 correlated roles as 51 fresh randomization revelations;
4. separate proper-prior e-process and Shiryaev–Roberts (SR) accumulators, preserving the distinction between lifetime crossing probability and average run length (ARL);
5. same-information rate, likelihood-surprise, covariance, CUSUM, and logistic controls; and
6. a detector/outcome freeze and a falsifiable conjunctive empirical gate.

Run 6 processes classical bits produced by fixed QEC measurements. It is not a quantum algorithm, a new measurement, or a claim of quantum speedup. It cannot beat a correctly specified same-information likelihood ratio in expected log growth, nor can it reveal physical changes that leave the archived syndrome law unchanged. These boundaries are part of the method, not qualifications added after results.

2 Observation Level, Related Work, and Evidence Boundary

2.1 Three observation levels

The experiment begins after the hardware measurement. Table 1 separates the three levels that must not be conflated.

Table 1: Observation levels and Run 6 access.

Level	Object and relevant theory	Run 6 access
Quantum acquisition	Unmeasured state/channel and measurement choice; Helstrom or quantum quickest-change limits [12, 13].	No.
Classical stream	Fixed detector bits in archive order; likelihood, CUSUM, covariance, and predictable betting.	Yes.
Retrospective utility	Observable flips and decoder predictions used for risk–coverage after detector freeze.	Outcome join only.

Classical-shadow sequential detection requires randomized measurement settings and a known inverse shadow channel [14, 15]. Those acquisition records are not present in the fixed syndrome archives. A numerical proxy would therefore compare different experiments and is excluded.

2.2 Sequential inference and change detection

Nonnegative test martingales and e-values provide anytime evidence under declared nulls [7, 8, 16]. Predictable betting gives concrete bounded-mean constructions [9]. E-detectors provide CUSUM- and SR-type ARL formulations [17]; they do not turn every SR threshold into a lifetime false-alarm probability. Sequential randomization e-values are also prior art [18]. Run 6 applies these ideas with one orientation coin per complete QEC shot.

The empirical controls deliberately span simple and structured alternatives: detector firing rate, Page's CUSUM [10], diagonal Bernoulli surprise, Hotelling geometry with Ledoit–Wolf shrinkage [11, 19], and a pairwise online logistic learner that receives the same 300 coordinates as the sparse witness. Published detector-likelihood work motivates, but is not identical to, the diagonal surprise control used here [20].

2.3 Evidence classes

We use four labels:

1. **Exact design audit**: complete observed pairs receive independently randomized orientation.
2. **Natural approximate event**: archived Google order contains an author-identified, approximately located event.
3. **Constructed boundary**: real-hardware PNNL/IBM snapshot cohorts are concatenated by the benchmark.
4. **Retrospective utility**: frozen detector ranks are joined to decoder mismatches.

Only the first label supports the declared conditional randomization guarantee. The other arms supply empirical evidence at frozen operating points.

3 Preregistered Data Roles and Experimental Unit

3.1 Google repetition-code replay

The primary archive is the Google Quantum AI deposit accompanying its surface-code study [1, 21]. The locked experiment `repetition_code_bZ_d25_r50_center_5_5` contains 500,000 shots, 50 syndrome rounds, and 1,224 declared detectors. Detector coordinates from the ideal Stim circuit define a permutation into 51 round roles by 24 physical check coordinates. The parser does not assume that a raw reshape preserves the physical order.

The immutable split is shown in Table 2. Held monitored shot $40000 + q$ is paired one-to-one with reference shot $20000 + q$, for $q = 0, \ldots, 19999$. Every complete pair consumes two archived physical shots. A role state is updated once per shot and is never shared with another role.

Table 2: Frozen zero-based Google shot split.

Indices	Role	Permitted use
$[0, 5000)$, $[10000, 15000)$	paired fit	model state
$[5000, 10000)$, $[15000, 20000)$	threshold	operating points
$[20000, 40000)$	held reference	one per test shot
$[40000, 60000)$	held monitor	primary replay
$[60000, 500000)$	future	untouched in primary

The primary event window is $[57750, 57800)$, with frozen narrow and wide sensitivity windows $[57770, 57780)$ and $[57725, 57825)$. The archive documentation localizes a large event only approximately; no exact physical onset or wall-clock delay is inferred.

3.2 PNNL/IBM auxiliary arm

The auxiliary data are the versioned PNNL release of IBM repetition-code experiments [22, 23]. Physical paths are reconstructed from each state-specific QASM. The independent summary unit is a path/snapshot, not an individual syndrome bit. Every comparison is called a *constructed boundary between real-hardware snapshot cohorts*. It is not interpreted as a natural time series; changed QASM is reported as circuit-and-hardware domain shift.

3.3 One randomization unit, one formal time

For complete paired shot u, a single random bit swaps its A/B orientation. The same bit acts on all roles $r = 0, \ldots, 50$. All role scores use pre-shot states, after which the 51 role-component experts are updated in parallel. Formal time is u, not $51u + r$. The flattened cycle index is retained only for empirical alert localization.

Frozen causal sequence
fit pairs → checkpoint clone → disjoint threshold replay → held detector replay → hash detector artifacts → outcome join
One complete pair → 51 simultaneous role scores → one formal update.

Figure 1: Protocol timeline. Threshold-state updates do not flow into held monitoring, and outcome labels cannot affect detector scores or ties.

4 S-PACE Construction

4.1 Bounded global, sparse, and spectral representations

For one cycle let $e \in \{0, 1\}^{24}$, $z = \mathbf{1} - 2e$, and

$$g(e) = \frac{1}{24} \sum_{j=1}^{24} e_j.$$

The fixed sparse feature bank is

$$\phi(e) = \left(e_1, \ldots, e_{24}, \left\{ \frac{1 + z_i z_j}{2} \right\}_{1 \leq i < j \leq 24} \right) \in [0, 1]^{300}. \tag{1}$$

All 276 check pairs are included, preventing an event-dependent adjacency choice. The spectral representation is

$$R(e) = \frac{zz^\mathsf{T}}{24}, \qquad R(e) \succeq 0, \quad \operatorname{Tr} R(e) = 1. \tag{2}$$

For paired reference/monitor cycles,

$$D^{\phi}_{u,r} = \phi(e^B_{u,r}) - \phi(e^A_{u,r}), \qquad \Delta_{u,r} = R(e^B_{u,r}) - R(e^A_{u,r}). \tag{3}$$

The separately designated M0 control uses $g(e^B) - g(e^A)$. It is not a component of the fixed proposed method S, which is exactly the M4/M5 composite defined below. Reporting M0 alongside S supplies a directly interpretable global-rate baseline; a pure orbit-centered relative representation would be pathwise blind to an additive common mode, as Sec. 5 makes explicit.

4.2 Sparse analytical witness

For each role, sparse witnesses retain exponentially weighted, past-only contrasts with half-lives $h \in \{4, 16, 64, 256\}$. For $k \in \{1, 4, 16, 64\}$, the witness assigns weight $1/k$ to the signed copies of the k largest absolute historical contrasts, using a fixed magnitude/index tie rule. Its ℓ_1 norm is at most one, so the current score $s^{(4)}_{u,r,h,k} = w^{\top}_{u-1,r,h,k} D^{\phi}_{u,r}$ lies in $[-1, 1]$. Scoring precedes the EWMA update.

4.3 Spectral analytical-observable-contrast witness

For $h \in \{4, 16, 64\}$, let $\overline{\Delta}_{u-1,r,h}$ be the past-only EWMA operator contrast. The stored matrix is symmetrized in float64. Effects are recomputed at prior role-update counts $8, 16, 24, \ldots$ and first used on the following observation, with absolute eigentolerance $\tau_{\mathrm{eig}} = 10^{-10}$. At a locked refresh, the full effect is

$$E^{+}_{u-1,r,h} = \Pi_{+}(\overline{\Delta}_{u-1,r,h}; \tau_{\mathrm{eig}}), \tag{4}$$

where $\Pi_{+}(H; \tau) = \sum_{\lambda_i(H) > \tau} v_i v_i^{\top}$, or zero if the index set is empty. For the rank-one variant, if $\lambda_{\max} \leq \tau_{\mathrm{eig}}$ the effect is zero. Otherwise form the projector $P_{\max}$ of all eigenvalues within τ_{eig} of $\lambda_{\max}$; choose the smallest coordinate j whose diagonal entry is within tolerance of $\max_i (P_{\max})_{ii}$; and set

$$v = \frac{P_{\max} e_j}{\|P_{\max} e_j\|_2}, \qquad E^{(1)} = v v^{\top}.$$

The first entry of v with magnitude above tolerance is made positive, although its projector is sign invariant. Both effects obey $0 \preceq E \preceq I$; therefore

$$s^{(5)}_{u,r,h,\mathrm{rank}} = \mathrm{Tr}(E_{u-1,r,h,\mathrm{rank}} \Delta_{u,r}) \in [-1, 1]. \tag{5}$$

The positive-part variational identity is the standard Jordan/Helstrom calculation, applied here to an already observed classical feature matrix; it is not a physical Helstrom measurement.

4.4 Factors, fixed priors, and empirical score

Every bounded predictable score is converted to

$$L_{u,r,c} = 1 + \beta_c s_{u,r,c}, \qquad \beta_c \in \{0.1, 0.3, 0.6, 0.9\}. \tag{6}$$

M4 has 64 uniform components over (h, k, β); M5 has 24 over $(h, \mathrm{rank}, \beta)$. The exact S-PACE composite assigns one half of its prior to each branch and remains a fixed mixture. It never selects the better branch from held performance.

The empirically calibrated cycle statistic is separately fixed as

$$Z^{(S)}_{q,r} = \max\{Z^{(4)}_{q,r}, Z^{(5)}_{q,r}\}, \tag{7}$$

where each branch maximum ranges only over its predeclared components. This maximum is *not* called an e-factor; its full multiplicity is calibrated on the disjoint threshold block.

5 Guarantees and Impossibility Boundaries

5.1 Predictable factors

Proposition 1 (bounded predictable factor)**.** *Let $s_u \in [-1, 1]$ satisfy $\mathbb{E}_0[s_u \mid \mathcal{F}_{u-1}] = 0$, and let $\beta_u \in [-1, 1]$ be $\mathcal{F}_{u-1}$-measurable. Then $L_u = 1 + \beta_u s_u$ is nonnegative and $\mathbb{E}_0[L_u \mid \mathcal{F}_{u-1}] = 1$.*

Proof. The range assumptions give $L_u \geq 0$, and conditional linearity gives $\mathbb{E}_0[L_u \mid \mathcal{F}_{u-1}] = 1 + \beta_u \mathbb{E}_0[s_u \mid \mathcal{F}_{u-1}] = 1$. $\square$

This proposition is a direct predictable-betting specialization, not a new martingale theorem. It applies to the independently randomized orientation because an antisymmetric current-pair score is conditionally centered given the unordered pair and past state. Fixed natural hardware orientation is not made exchangeable by this notation.

5.2 Complete-shot expert mixture

Proposition 2 (shared-orientation roles)**.** *Suppose one orientation coin is drawn for complete paired shot u, and every role-component score is computed from its pre-shot state. If each (r, c) construction is a conditional e-factor with respect to the pre-coin filtration, then every fixed convex mixture of the corresponding role-component wealth processes is an e-process. Serially treating the 51 roles as fresh coin revelations is not justified.*

Proof. Each expert wealth is nonnegative with conditional expectation no larger than its previous value. Conditional linearity preserves this property under fixed convex weights despite arbitrary dependence between experts. After the first role reveals the shared orientation, the remaining roles do not receive a fresh randomization bit, so a 51-step product needs an additional conditional argument that the design does not supply. $\square$

For horizon H, the locked proper-prior recursion is

$$A_{0,r,c} = 0, \qquad A_{u,r,c} = L_{u,r,c}\left(A_{u-1,r,c} + \frac{1}{H}\right), \tag{8}$$

$$E_u = \frac{H-u}{H} + \frac{1}{51}\sum_{r=0}^{50}\sum_{c} \pi_c A_{u,r,c}. \tag{9}$$

Here $H = 20{,}000$ for held replay and $H = 5{,}000$ for the randomization threshold replay.

5.3 Common-mode blind spot and global diagnostic

Proposition 3 (relative-projection blind spot)**.** *Let $P_{\mathrm{rel}} = I - P_{\mathrm{triv}}$ remove the invariant spatial mode. If a change is exactly $\Phi'_u = \Phi_u + \mathbf{1} \otimes c_u$, with no clipping, nonlinear remapping, or relative change, then*

$$P_{\mathrm{rel}}\Phi'_u = P_{\mathrm{rel}}\Phi_u.$$

Every detector measurable only with respect to the relative field has the same path law before and after this change.

Proof. The additive term belongs to the invariant subspace and is annihilated by P_{rel}. Equality is pathwise, so equality of every relative-field functional follows. $\square$

This projection identity motivates reporting M0 as a separate, directly interpretable global-rate diagnostic beside the fixed M4/M5 composite S. M0 is a control, not a claimed repair or an additional branch of S, and the identity is not a new statement about topological order.

5.4 Closed-form maximum-difference coordinates

Proposition 4 (top-k capped contrast)**.** *For mean contrast $\delta \in \mathbb{R}^p$ and $k \in \{1, \dots, p\}$, let $v = (\delta, -\delta)$ and*

$$\mathcal{C}_k = \left\{ q \in \mathbb{R}_+^{2p} : \sum_{i=1}^{2p} q_i = 1,\ \ q_i \le \frac{1}{k} \right\}.$$

Then

$$\max_{q \in \mathcal{C}_k} q^\mathsf{T} v = \frac{1}{k} \sum_{j=1}^{k} |\delta|_{(j)}. \tag{10}$$

Proof. A linear functional over the capped simplex assigns the largest allowed mass to its largest coordinates. The k largest coordinates of $(\delta, -\delta)$ are the correctly signed copies of the k largest absolute entries of δ. □

Equation (10) solves the declared linear-gap objective. It does not prove minimum detection delay or universal feature-selection optimality.

5.5 Spectral gap and likelihood ceiling

Proposition 5 (positive-part operator gap)**.** *For Hermitian $\overline{\Delta}$,*

$$\max_{0 \preceq E \preceq I} \operatorname{Tr}(E\overline{\Delta}) = \operatorname{Tr}(\overline{\Delta}_+),$$

attained by the unambiguous spectral projector $E = \Pi_+(\overline{\Delta}; 0)$ onto the strictly positive eigenspace.

This standard variational identity maximizes a one-step linear gap, not sequential log wealth. Moreover, if $P_1 \ll P_0$ is a correctly specified same-information alternative, every one-step e-factor L obeys

$$\mathbb{E}_1 \log L \le \mathrm{KL}(P_1 \| P_0), \tag{11}$$

with equality for the likelihood ratio dP_1/dP_0. Indeed,

$$\mathbb{E}_1 \log L = \mathrm{KL}(P_1 \| P_0) + \mathbb{E}_1 \log\left(L \frac{dP_0}{dP_1} \right) \le \mathrm{KL}(P_1 \| P_0)$$

by Jensen and $\mathbb{E}_0 L \le 1$. Thus potential benefits of S-PACE concern robustness, transparent calibration, sparse interpretation, or model-fitting cost—not violation of the likelihood ceiling [24].

5.6 Lifetime control versus average run length

Theorem 6 (declared sequential meanings)**.** *Under the exact conditional null, Eq. (9) is a nonnegative supermartingale with $E_0 = 1$, so*

$$\mathbb{P}_0\left(\sup_{u \le H} E_u \ge 1/\alpha \right) \le \alpha. \tag{12}$$

Separately, if

$$R_{u,r,c} = (R_{u-1,r,c} + 1) L_{u,r,c}, \qquad R_u = \frac{1}{51} \sum_{r,c} \pi_c R_{u,r,c}, \tag{13}$$

then the SR stopping rule at threshold γ has null ARL at least γ complete paired shots*, subject to the standard optional-sampling conditions. A role or flattened cycle index can localize an empirical alert within a shot but is not an ARL time unit.*

These statements are not interchangeable. When $L \equiv 1$, the SR statistic advances like a clock and crosses a finite threshold. Run 6 uses $\alpha = 0.01$ for the reported proper-prior audit. Equation (13) records the distinct SR/ARL construction and its formal unit; this manuscript does not report a separate Google SR result.

5.7 Accessible-data no-go

Theorem 7 (pushforward no-go)**.** *Let Y denote the archived syndrome path and let $T(Y, U)$ be any detector, stopping rule, or randomized post-processing, where auxiliary randomness U has the same law under both hypotheses. If the two physical regimes induce the same law for Y, they induce the same law for $T(Y, U)$.*

Proof. The result is the data-processing identity for the pushforward of the common law of (Y, U). □

Theorem 7 does not discover Wilson loops, toric-code logical structure, or a string-theory result. Wilson observables and toric-code strings are established physical constructions [25, 26]. They are not measured in the primary repetition-code archive.

6 Locked Comparators and Evaluation

6.1 Methods and information parity

Table 3 records the frozen method set. M0C, M1, and M2 are mandatory contextual controls but cannot rescue failure of the predesignated M0/M3 comparisons.

Table 3: Frozen methods. "Exact" refers only to the randomized complete- pair design with the displayed factor construction.

ID	Score or fitted object	Status
M0	paired detector firing-rate difference	exact-factor/empirical
M0C	within-shot two-sided Page–CUSUM	empirical
M1	fixed diagonal Bernoulli surprise	exact-factor/empirical
M2	role-centered shrinkage Hotelling score	empirical
M3	past-only pairwise logistic, same ϕ	exact-factor/empirical
M4	sparse signed top-k witness	exact-factor/empirical
M5	positive-eigenspace/rank-one witness	exact-factor/empirical
S	fixed M4/M5 prior; empirical max score	as declared above

M0 and M0C. With $d^g_{q,r} = g(e^B_{q,r}) - g(e^A_{q,r})$, M0 uses $s^{(0)}_{q,r} = d^g_{q,r}$, eight signed bets $\beta \in \{\pm 0.1, \pm 0.3, \pm 0.6, \pm 0.9\}$, and empirical score $Z^{(0)}_{q,r} = |s^{(0)}_{q,r}|$. At shot q, define the M0C channels $x_{r,0} = d^g_{q,r}$ and $x_{r,j} = e^B_{q,r,j} - e^A_{q,r,j}$ for raw checks $j \in \{1, \ldots, 24\}$. M0C resets at each shot and, for drifts $\kappa \in \{0.01, 0.05, 0.1\}$, applies

$$C^+_{\kappa,c} \leftarrow \max\{0, C^+_{\kappa,c} + x_{r,c} - \kappa\}, \qquad C^-_{\kappa,c} \leftarrow \max\{0, C^-_{\kappa,c} - x_{r,c} - \kappa\}.$$

Its score is $\max_{\kappa,c}\{C^+_{\kappa,c}, C^-_{\kappa,c}\}$; it is a within-shot empirical scan, not an e-factor or cross-shot quickest-change oracle.

M1 and M2. M1 freezes role/check probabilities from the fit pairs using Jeffreys smoothing and clipping,

$$\widehat{p}_{rj} = \operatorname{clip}\left(\frac{c_{rj} + 1/2}{10001}, 10^{-4}, 1 - 10^{-4}\right),$$

forms the mean diagonal Bernoulli negative log likelihood $\ell_r(e)$, and scores

$$s_{q,r}^{(1)} = \frac{\ell_r(e_{q,r}^B) - \ell_r(e_{q,r}^A)}{\log\{(1 - 10^{-4})/10^{-4}\}} \in [-1, 1].$$

M2 uses the exact PCG64(610601) role-stratified 20,000-observation fit subsample, role means $\widehat{\mu}_r$, and one pooled symmetrized Ledoit–Wolf precision $\widehat{\Omega}$:

$$Z_{q,r}^{(2)} = \max\left\{0, (D_{q,r}^{\phi} - \widehat{\mu}_r)^{\mathsf{T}} \widehat{\Omega} (D_{q,r}^{\phi} - \widehat{\mu}_r)\right\}.$$

M2 is empirical and never inherits an e-process guarantee.

M3 same-feature control. For $\eta \in \{0.001, 0.01, 0.1\}$, M3 maintains a distinct $w_{r,\eta} \in \mathbb{R}^{300}$, no intercept, and L2 coefficient $\lambda = 10^{-4}$. From the pre-update $a = w_{r,\eta}^{\mathsf{T}} D_{q,r}^{\phi}$ and $2c = \max\{1, \|w_{r,\eta}\|_1\}$, it scores

$$s_{q,r}^{(3,\eta)} = \tanh(a/(2c))$$

and only then applies the overflow-safe pairwise-logistic update

$$w_{r,\eta} \leftarrow w_{r,\eta} + \eta\, \sigma(-a) D_{q,r}^{\phi} - \eta \lambda w_{r,\eta}.$$

The empirical maximum over the three frozen experts is threshold-calibrated; the exact branch instead uses its fixed 12-component (η, β) mixture.

All methods receive the same raw detector records. The 5,000-pair fit block is common. The disjoint 5,000-pair threshold clone selects the smallest observed-or-infinite strict threshold whose replay emits at most two notifications among 255,000 cycle-role opportunities. At most one notification is counted per shot; states continue updating through within-shot suppression. The complete threshold frontier is frozen before held scoring.

6.2 Primary event and risk–coverage endpoints

The natural-event endpoints are: alarm inside each frozen event window, first alarm shot and role, pre-event alerts, and event score rank. Because the event is a single cluster with approximate onset, cycle-level population confidence claims are prohibited.

Only after detector artifacts are hashed are actual observable flips and decoder predictions opened. At the primary triage budget, each method ranks the 20,000 monitor shots by frozen score, breaking ties by ascending archive index. The primary label is actual observable flip XOR correlated-matching prediction; PyMatching is secondary. The top 20 shots are selected without labels. This is retrospective veto/triage, not a new decoder.

6.3 Post-detector execution repair, post-outcome publication repair, and provenance

The detector implementation, manifest, and original ratification were committed and pushed before detector-value access. The detector replay then read, scored, and serialized the held detector stream. The first randomization launch stopped in artifact-schema validation: all 32 planned shards produced the same failure, no shard manifest and no completed replicate. Diagnosis exposed detector-derived numeric diagnostics.

The defect was a producer–consumer validation mismatch: the producer omitted one redundant SR metadata key, while the consumers required it, and the consumer's expected prior vector did not reproduce the producer's exact normalization path. The repair was fixed from source and schema evidence. No detector-derived number selected the repair; no threshold, score, state, seed, window, endpoint, or claim gate changed. The original 231-artifact detector bundle was reused byte-for-byte and was not rerun.

A second, post-detector and pre-outcome implementation–manifest–ratification chain binds the exact allowlisted consumer and verification diff, the original ratification, detector manifest and artifact registry, 32 preserved failed attempts, environment, thread settings, package lock, and runtime module origins. Its access record truthfully sets detector access and numeric diagnostic exposure to true, repair selection from those numbers to false, decoder-outcome and PNNL held access to false, and completed randomization to zero. Randomization, PNNL, the outcome join, and the final decision must bind this repair ratification; the immutable detector manifest continues to bind only the original ratification.

Thus the methods and gates were initially preregistered, but the end-to-end execution is not described as detector-blind repair or pristine full preregistration. This chronology is a limitation and part of the machine-checked publication contract.

After every locked scientific producer, including the outcome join, had completed, the first strict publication extraction stopped before emitting a bundle because its consumer treated the PNNL adaptive-state resource ledger as an object although the frozen producer emits an ordered 22-row array. The repair changed only the publication extractor, its tests, and provenance documentation. The corrected consumer now checks exact cohort/state order, row schema, path dimension, role count, all five accumulator dimensions, and the independently reconstructed numeric-byte identities. It did not alter or rerun a scientific producer and changed no upstream artifact, threshold, seed, endpoint, gate, outcome, or recorded runtime. The failed consumer had not reached the outcome manifest, but the outcome stage already existed; therefore this amendment is described neither as outcome-blind nor as preregistered. Separate publication-provenance records preserve the failure, the exact allowlisted repair, and the unchanged hashes of all eight experimental evidence inputs.

6.4 Locked Google uncertainty

Risk–coverage stability uses a paired circular moving-block bootstrap of complete monitored shots: block length 128, 2,000 replicates, and replicate seed $611000 + b$. Each resample draws 157 circular block starts, truncates to 20,000 shots, resamples score/outcome rows together, and recomputes the fixed top-B metrics and differences. A separate descriptive threshold bootstrap uses complete threshold shots, block length 128, 2,000 replicates, 40 block starts, and seeds $613000 + b$. Neither bootstrap replaces the deterministic threshold or changes the primary Boolean. With one natural event cluster, neither interval is called a population-level event-power confidence interval.

6.5 Exact randomization audit

The threshold block is replayed under 256 predeclared random-swap seeds. One orientation bit applies to all 51 roles in a complete pair, all states reset between replicates, and factors are mixed only at complete-shot time. This arm audits implementation and design-based calibration. It neither establishes natural hardware stationarity nor estimates event power.

6.6 PNNL path-level uncertainty

Each of the 22 path–logical-state streams calibrates its empirical threshold with 4,096 circular moving-block bootstrap replicates of *complete paired shots*, block length 32, and its frozen cohort/state/method seed. This within-stream threshold bootstrap is not the cross-cohort uncertainty unit.

For the auxiliary comparison, logical states 0 and 1 are first averaged within each of the 11 physical path/snapshot pairs. The primary uncertainty analysis then bootstraps these 11 paired path effects 10,000 times with PCG64 seed 612500. A separate 10,000-draw sensitivity bootstrap clusters by the five ordered calibration-hash pairs with seed 612501. The paired randomization analysis exhaustively enumerates all 2^{11} sign flips of the 11 path effects. No bootstrap or randomization treats individual syndrome bits as independent.

The exact swap audit is a different operation: seeds 610700 through 610955 create one A/B swap bit per complete paired shot, shared across all methods and round roles; every path–state replicate resets fully, scores before updating, and performs one role-expert mixture update per shot. It validates the design, not natural IBM drift. QASM equality and inequality remain visible so circuit-and-hardware domain shift cannot be mislabeled as hardware-only drift. The publication consumer validates all 110 ordered path–state–method JSON rows and aggregate-cross-checks their alarm totals and global maxima against hash-bound 256×5 count and maximum-log-e arrays. Those aggregate arrays do not reconstruct the allocation of alarms among individual path rows, so the audit is not described as an independent row-level recomputation.

6.7 Locked conjunctive empirical gate

Let G be `google_primary_pass`. It is true only if the fixed composite:

1. alarms inside $[57750, 57800)$ at the validation-locked primary threshold;
2. has at most nine pre-event alerts;
3. was frozen with its resource ledger before outcome access;
4. captures at least one more top-20 correlated-decoder mismatch than paired DFR;
5. captures at least one more than same-feature online logistic;
6. reports every frozen sensitivity and uncertainty summary; and
7. receives no extra records or outcome labels.

Let P be the single aggregate `pnnl_retention_pass`. Against *each* of DFR and same-feature online logistic, the fixed composite must have no more pre-boundary false alarms over all 22 state streams and a strictly lower equal-weight macro restricted-delay fraction after logical-state averaging. Ties fail. This is not a count or selection of "retained paths"; all 11 path rows and both logical-state summaries remain in the report. The overall statement is the conjunction

$$C_{\mathrm{Run}\,6} = G \wedge P. \tag{14}$$

If either gate fails, the required conclusion is exactly "no demonstrated S-PACE algorithmic advantage." A tie, a sparse heat map, or a successful randomization audit cannot override Eq. (14).

6.8 Physical and computational resource ledger

Observation exposure is reported for every method: calibration/reference and monitor shots, QEC cycles, detector bits, feature dimension, and update frequency. Offline pairing consumes two physical archived shots per formal update. The implementation executes all methods in one detector runner, so wall time and maximum resident set size are identifiable only for the *joint pipeline/process*. They are not divided into unsupported per-method timing or RSS claims. The process-wide ledger additionally records output bytes and high-level operation counts where the manifest supports them. Formal e-process/ARL time is always a complete paired shot; role and cycle indices are only within-shot localization coordinates.

7 Locked Results and Dual Provenance

Pre-results status. Every red `TBD(KEY)` marker is a conditional fallback. Final mode accepts only fragments generated from the original freeze, repair chain, and four completed hash-bound result manifests by the fail-closed publication extractor; editing this manuscript cannot resolve a marker.

7.1 Provenance and detector freeze

The publication bundle verifies three provenance layers. The immutable detector manifest binds the original pre-access ratification. The randomization, PNNL, outcome, and decision records additionally bind the post-detector execution-repair ratification and its truthful access record. The eight experimental evidence inputs retain their original hashes and chronology. Separate post-outcome publication-consumer records bind the preserved failed extraction and the exact allowlisted reporting repair; no result manifest is retroactively rewritten to bind them. The extractor rejects any attempt to describe the repair as detector-blind, any detector rerun, or any completed randomization or held-outcome/PNNL access before repair. It verifies every derived artifact digest and recomputes both empirical gates before emitting the claim sentence. The bundle contract is `TBD(PUBLICATION_BUNDLE_MANIFEST)`. Detector-only artifacts precede both the repair ratification and the separately hash-authorized outcome join.

7.2 Threshold frontier and locked uncertainty

Table 4: Validation-locked thresholds and complete alert frontier. Strict crossing and one-notification-per-shot semantics remain fixed.

`TBD(GOOGLE_THRESHOLD_FRONTIER_TABLE)`

Table 5: Descriptive Google threshold-bootstrap audit for all methods. Complete-shot circular blocks have length 128; all 2,000 replicates and seeds 613000–614999 are validated. The summaries do not replace a frozen threshold, enter a locked Boolean, or estimate natural-event power.

`TBD(GOOGLE_THRESHOLD_BOOTSTRAP_TABLE)`

Table 6: Locked Google S-PACE-minus-comparator uncertainty for both labels, both comparators, budgets $2, 20, 200$, and all four metrics, including valid replicate counts. Resampling uses complete shots/blocks and does not change the deterministic primary Boolean; only primary-label budget-20 capture is gate-relevant.

TBD(GOOGLE_UNCERTAINTY_TABLE)

Table 7: Frozen Google event results for all predeclared methods, including the mandatory CUSUM, diagonal-surprise, and Hotelling controls.

TBD(GOOGLE_EVENT_TABLE)

7.3 Google natural-event replay

7.4 Frozen decoder-outcome join

7.5 Randomization-design calibration

7.6 PNNL/IBM constructed-boundary arm

7.7 Resource ledger

7.8 Locked decision

The result-contingent sentence emitted from that recomputation is: TBD(CLAIM_SENTENCE)

8 Discussion

8.1 What a satisfied conjunctive gate would and would not mean

If and only if both gates in Eq. (14) pass, the supported statement is an empirical fact: both fixed conditions were satisfied for these datasets, implementations, endpoints, and budgets. Because the tests compare specific implementations at fixed endpoints, passing does not establish superiority over same-feature or same-parity logistic or threshold classes, a correct likelihood or oracle rule, or a general algorithmic advantage. It also does not establish universal sample efficiency, scaling advantage, or quantum acceleration. If either gate fails, the benchmark remains informative by identifying the simpler comparator that matched or exceeded the composite without changing the success criterion.

8.2 Symmetry and physics language

Sector projections are classical representation bookkeeping on observed features. Group-symmetric quantum hypothesis testing [27] and symmetry-resolved entanglement [28] are related mathematical contexts, not Run 6 outputs. The primary data are a one-dimensional repetition-code syndrome archive; they do not constitute a Wilson-loop measurement, a new topological-order theorem, or evidence about string theory or holography.

8.3 Limitations

The primary hardware evidence contains one author-identified event cluster. Its onset is approximate, and archived shot order is not a calibrated wall-clock cadence. The paired reference reservoir is historical rather than physically simultaneous, so exact exchangeability is supplied only by the

TBD(GOOGLE_EVENT_FIGURE)

Figure 2: Google natural-event causal replay. The author-provided event location is approximate; the plot does not define an exact onset.

Table 8: Risk–coverage results from frozen rankings for both decoder labels and all locked budgets $2, 20, 200$, with all eight methods, captured/total, recall, precision, retained mismatch rate, coverage, intervals, and valid replicate counts visible. Only correlated-matching capture at budget 20 enters the primary gate.

TBD(GOOGLE_RISK_BUDGET_TABLE)

artificial randomization audit. PNNL comparisons are constructed snapshot boundaries and some confound circuit with hardware changes. The same 300-coordinate bank is broad but finite; no monitor can detect a change absent from its accessible syndrome sigma-field. The consumer validator was repaired only after detector values had been produced and numeric diagnostics had been exposed. Although the repair was source/schema-derived, left the detector bundle unchanged, and preceded outcome and held-PNNL access, this precludes describing the complete execution as detector-blind or pristine end-to-end preregistration. The post-outcome publication repair is a reporting-pipeline amendment, not additional experimental evidence. Its safeguards are the preserved first failure, immutable upstream hashes, an allowlisted diff, and fresh recomputation of the unchanged gates. Finally, the full spectral branch costs repeated eigendecompositions, so any computational benefit must be measured rather than inferred from parameter count.

8.4 Practical next experiment

The most consequential follow-up is an online surface-code experiment with prospectively timestamped calibration interventions and synchronized decoder likelihoods. A resource-matched protocol should compare S-PACE with covariance/CUSUM, detector likelihood, adaptive decoder/noise trackers, and predeclared gauge-invariant diagnostics at identical shots, qubit-cycle exposure, and intervention cost. Classical-shadow e-detection becomes a valid comparator only if its randomized acquisition settings are collected as part of that separate experiment.

9 Conclusion

This work specifies an auditable route from past-only maximum-difference features to sequential monitoring of fixed QEC syndrome records. Its exact content is deliberately narrow: bounded predictable factors under a declared conditional design, fixed mixtures at the complete-shot experimental unit, and standard lifetime/ARL guarantees used without interchange. The result-contingent conclusion is: TBD(CLAIM_SENTENCE) The benchmark is designed so that a strong rate, covariance, likelihood, or same-feature comparator remains a scientific result rather than a baseline to be removed after inspection.

TBD(GOOGLE_RISK_COVERAGE_FIGURE)

Figure 3: Retrospective utility of detector-only frozen rankings.

Table 9: Complete-shot proper-prior randomization audit. No SR result is inferred or reported.
TBD(GOOGLE_RANDOMIZATION_TABLE)

Data and Code Availability

The Google and PNNL data are public deposits [21, 23]. The accompanying repository stores the initial preregistration, original freeze ratification, separately ratified post-detector repair, detector/outcome separation, truthful access record, environment lock, source hashes, and machine-readable resource ledgers. Final manuscript values and figures are accepted only through the three-layer-provenance, hash-bound publication bundle.

Acknowledgments

The author thanks the creators of the public QEC datasets and the developers of the statistical and QEC tools on which this reproducible benchmark depends. This working paper makes no claim of endorsement by those groups.

A Frozen Hyperparameters

B Proof Detail for the Proper-Prior Process

Let $\rho_u = 1/H$ for $u = 1, \ldots, H$ and $\rho_{>u} = (H - u)/H$. From Eq. (8) and conditional e-validity,

$$\mathbb{E}_0[A_{u,r,c} \mid \mathcal{F}_{u-1}] \le A_{u-1,r,c} + \rho_u.$$

After fixed mixing,

$$\mathbb{E}_0[E_u \mid \mathcal{F}_{u-1}] \le \rho_{>u} + \frac{1}{51} \sum_{r,c} \pi_c (A_{u-1,r,c} + \rho_u) = E_{u-1}.$$

The initial tail mass gives $E_0 = 1$; Ville's inequality yields Eq. (12). Replacing the proper start prior by an unnormalized SR clock changes the guarantee and cannot retain the same lifetime interpretation.

C Result-Insertion Checklist

Before a submission build, all of the following must hold:

TBD(GOOGLE_RANDOMIZATION_FIGURE)

Figure 4: Exact-design randomization audit.

Table 10: PNNL exact paired-swap implementation audit. The table covers all 22 path–state episodes, five methods, and 256 seeds. It is not a natural-hardware null, enters neither empirical gate nor the overall conjunction, and cannot override a failed empirical gate.

TBD(PNNL_RANDOMIZATION_AUDIT_TABLE)

1. every keyed pending marker is absent from the compiled PDF;
2. the hash-bound publication bundle is independently regenerated from all eight declared experimental evidence inputs and the separate publication-repair provenance records, and matches every included byte;
3. the immutable detector binds the original pre-access ratification, while every downstream arm and the final decision bind the separately ratified repair;
4. the repair access record discloses diagnostic exposure and rejects detector-blind or pristine end-to-end preregistration language;
5. the post-outcome publication-repair records preserve the first failed extraction, bind an allowlisted consumer-only diff and unchanged upstream hashes, and reject outcome-blind language;
6. the abstract calls the Google onset approximate and PNNL a constructed snapshot boundary;
7. exact e-validity is confined to the randomized complete-pair design;
8. all 51 roles remain one formal time step;
9. e-process lifetime control and SR ARL are not conflated;
10. paired DFR and same-feature logistic both appear in the gate;
11. M0C, M1, and M2 are reported regardless of outcome;
12. the complete threshold/frontier and locked uncertainty tables are included, including the descriptive 2,000-replicate threshold bootstrap;
13. PNNL pre-false-alarm, miss, macro-delay, and both comparator retention conditions are included without selecting paths, together with the complete paired-swap implementation-audit summary;
14. the recorded chain shows zero completed randomization replicates and no outcome or held-PNNL access before repair, with no detector rerun;

Table 11: PNNL equal-weight macro summaries. Pre-boundary false-alarm and miss counts cover all 22 path–state streams; no path is selected or discarded by the retention decision.

TBD(PNNL_MACRO_PREFA_MISS_TABLE)

Table 12: PNNL state-resolved results. Each cell is pre-boundary false-alarm indicator / miss indicator / restricted post-boundary delay fraction. All 22 ordered path–state rows and all five methods are reported; no path or logical state is selected.

TBD(PNNL_STATE_RESULTS_TABLE)

15. uncertainty is clustered by shot/event or path as declared;
16. a tied or failed gate produces the locked negative conclusion; and
17. no quantum speedup, universal sample-efficiency, topological, string-theory, or holographic claim appears.

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Table 13: PNNL state-averaged cohort results. Each cell is pre-false-alarm mean / miss mean / restricted post-boundary delay fraction after averaging logical states 0 and 1. All 11 cohorts and all five methods are reported; the forest plot separately shows the two gate-relevant differences.

TBD(PNNL_COHORT_RESULTS_TABLE)

Table 14: Aggregate PNNL retention conditions against both predesignated comparators, with the path bootstrap, calibration-pair sensitivity bootstrap, exact paired sign-flip p-value, and all three gate Booleans.

TBD(PNNL_AGGREGATE_RETENTION_TABLE)

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Table 15: PNNL cohort-control map. The anonymous C01–C11 plot labels map to the frozen path/snapshot cohorts here; QASM equality is exposed, and the control class is derived per cohort from the hash-bound Pittsburgh manifest. These constructed snapshot comparisons are not natural drift.

TBD(PNNL_COHORT_CONTROL_TABLE)

TBD(PNNL_DELAY_FOREST_FIGURE)

Figure 5: PNNL/IBM constructed snapshot-boundary results.

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Table 16: Hash-bound resource ledger. Timing and maximum RSS are process-wide measurements for the joint pipeline and are not attributed to individual methods. All three Google joint replays, both PNNL wall-time scopes, and PNNL bootstrap/randomization surrogate work and eigendecompositions are reported; surrogate work is not new hardware exposure.

TBD(JOINT_PIPELINE_RESOURCE_LEDGER)

Table 17: Recomputed atomic predicates and conjunctive locked decision.

TBD(RECOMPUTED_GATE_DECISION_TABLE)

Table 18: Selected locked parameters.

Object	Frozen value
M3 learning rates	$0.001, 0.01, 0.1$; L2 10^{-4}, no intercept
M4 half-lives	$4, 16, 64, 256$ role updates
M4 supports	$k = 1, 4, 16, 64$
M5 half-lives	$4, 16, 64$ role updates
M5 effects	stride 8; $\tau_{\text{eig}} = 10^{-10}$; deterministic anchored rank one and full positive eigenspace
Bet fractions	$0.1, 0.3, 0.6, 0.9$
Proper-prior threshold	$\alpha = 0.01$, crossing at 100
Empirical cycle budget	at most 2 notifications in 255,000 opportunities
Risk budgets	top $2, 20, 200$ of 20,000 monitor shots
Google randomization	256 predeclared complete-shot swap streams
Google threshold uncertainty	2,000 shot-block draws, block length 128
PNNL stream threshold	4,096 shot-block draws, block length 32
PNNL paired-swap audit	256 seeds; 22 path–state episodes; five methods
PNNL uncertainty	10,000 path draws; 10,000 cluster sensitivity draws; all 2^{11} paired signs